

1. Executive Overview & The Luxury Spatial Commerce Challenge

Autonomous AI Sales & Spatial Design Advisor — Powered by Deterministic Guardrails & 2-Stage RAG

0ms / 0 Tokens	100% Deterministic	15" & 21-30"	100% Offline Edge
Fast-Path Router Bypass	Exact 18% GST & Math	Building Code Clearances	Local Ollama Appliance
Instant greetings & catalog tables	Zero mental math hallucinations	Centerline & front space rules	Zero cloud API dependency/fees

THE LUXURY SPATIAL COMMERCE CHALLENGE • Spatial Hallucinations & Clearance Violations: Pure LLMs lack spatial awareness, frequently recommending 6ft freestanding bathtubs for compact 4x5 ft powder rooms without verifying door swings or mandatory 15" toilet centerlines. • Pricing & Tax Arithmetic Inaccuracies: Multi-item luxury quotes involving 15% package discounts, tiered discounts, and 18% GST in Indian Rupees (INR) consistently trigger arithmetic hallucination errors in open-ended LLMs. • High Latency & Paid Cloud Token Waste: Traditional chatbot loops incur 3-6 second delays and paid API tokens for basic conversational greetings or repetitive full-catalog portfolio listings. • Catalog Drift & Out-of-Scope Hallucinations: Unconstrained models invent non-existent finishes or recommend out-of-catalog items (e.g. refrigerators) instead of adhering strictly to verified luxury inventory.	OUR AGENTIC SOLUTION & CORE PARADIGM • Decoupled Dual-Engine Architecture: Pairs an intelligent ReAct Orchestrator (LlamaIndex) with four deterministic Python rule, spatial, math, and logistics engines. The LLM directs intent; Python guarantees mathematical truth. • 0ms Fast-Path Intent Routing: Deterministic regex classifier intercepts casual greetings, catalog overviews, and single-spec queries instantly with zero token consumption and zero latency overhead. • Spatial Studio & Building Code Validator: Computes exact fixture capacities (compact powder room to master bath) and verifies plumbing building codes (15" centerline, 21"-30" front clearance, 30" door arc swing). • Native Indian Luxury Market Customization: Strictly binds all pricing to Indian Rupees (INR) with trailing 999 luxury pricing, 18% GST computations, and PIN-code based warehouse fulfillment lead times.
CORE ARCHITECTURAL PARADIGM: By replacing unconstrained generative responses with a Deterministic Rule Engine + 2-Stage Hybrid RAG + Autonomous ReAct Orchestrator, the platform guarantees 100% factual accuracy, zero arithmetic errors, and rapid sub-second consultative sales interactions.	

2. System Architecture & End-to-End ReAct Workflow

Multi-Stage Orchestration: Fast-Path Intent Router -> Autonomous Agent -> 4 Deterministic Rule Engines

1. Customer Query Text or Studio Input	2. Fast Router (0ms) Bypass Greetings / Catalog	3. ReAct Orchestrator LlamaIndex AgentWorkflow	4. 4 Deterministic Engines Spatial RAG Math Stock	5. Executive Proposal 18% GST + INR Quote + UI
1. Spatial Studio & Layout Engine tools/room_planner.py & bundle_optimizer.py • Clearance Enforcement: 15" toilet centerline, 21"-30" front basin/commode clearances, and 30" door arc swings. • Capacity Solver: Sizing limits for powder rooms (2 fixtures), guest baths (3), master baths (4), and luxury spas (5+). • Curated Themes: Modern Minimalist, Classic Luxury, Japanese Zen, Mid-Century, Smart High-Tech.		2. 2-Stage Local Hybrid RAG tools/catalog.py (ChromaDB + Cross-Encoder) • Stage 1 Dense Retrieval: Local ChromaDB vector database powered by BAAI/bge-small-en-v1.5 embeddings. • Stage 2 Neural Reranking: ms-marco-MiniLM-L-6-v2 Cross-Encoder reranks top candidates for exact SKU & finish matches. • In-Memory LRU Cache: Guarantees 0ms repeated retrievals on frequent queries.		
3. Precision Pricing & Tax Engine tools/calculator.py (Strict Python Arithmetic) • Itemized MRP Math: Computes exact catalog subtotals in Indian Rupees (INR 1 USD = 83.50 INR normalized). • Automated Bundle Savings: Applies verified 15% package discount calculations across multi-fixture suites. • Mandatory 18% GST Rollup: Computes exact tax subtotals, savings amounts, and final grand totals.		4. Warehouse Logistics & Stock Engine tools/inventory.py (Regional Dispatch Centers) • Real-Time Availability: Tracks inventory levels (In Stock, Low Stock) across primary distribution hubs. • PIN/ZIP Code Routing: Resolves closest regional dispatch (Mumbai, Delhi, Bengaluru) and calculates transit days (2-7 days). • Commercial Assurance: Injects manufacturer warranty and 30-day return policy into client proposals.		
HIGH RESILIENCY & ZERO-LAG CLOUD TIMEOUT FALLBACK: If cloud API latency exceeds 3.0s or rate limits trigger (HTTP 429), the system deterministically compiles a complete, formatted sales proposal directly from verified tool outputs in <50ms. Zero UI freeze, 100% uptime.				

3. Technical Stack & Dual-Runtime Deployment Matrix

Engineered for Zero-Cost Edge Hardware or Scalable Multi-Tenant Cloud Deployment

Architecture Layer	Technology & Component	Key Technical Capabilities & Engineering Rationale
Agent Orchestration	LlamaIndex v0.11+ AgentWorkflow & ReAct Loop	Event-driven ReAct state machine managing dynamic tool dispatch, multi-step reasoning, context retention, and strict sales persona prompt injection.
Vector Database	ChromaDB chromadb.PersistentClient	Persistent embedded local vector store with self-bootstrapping indexer (0-dependency cold startup on fresh machine).
Neural Embeddings & Reranker	BGE-Small-v1.5 + ms-marco-MiniLM-L-6-v2	Local HuggingFace embeddings combined with Cross-Encoder reranker and in-memory LRU cache for high-precision, 0ms repeated retrievals.
Dual LLM Engine	Ollama (Local Edge) / OpenRouter (Cloud API)	100% offline edge execution via qwen2.5:3b (Flash Attention) with seamless automatic fallback to cloud models if local runtime is absent.
Interactive Studio UI	Gradio 6.0 Custom UI	Interactive Spatial Studio 3D visualizer, responsive clearance sliders, quick-action chips, dark/light theme, and live sales advisor chat.
DevOps & Testing	Docker Compose + Scripts + 12-Scenario Test Suite	Containerized Docker deployment, Windows .bat / Linux .sh one-click launchers, and automated end-to-end verification suite.

MODE A: 100% OFFLINE EDGE APPLIANCE (SHOWROOMS)

- Runs locally on showroom workstation or laptop with consumer GPU / CPU.
- Zero API Costs:** 100% free token usage via quantized Ollama weights.
- Air-Gapped Privacy:** Customer architectural floor plans and quotes never leave the device.

MODE B: MULTI-TENANT CLOUD CLUSTER

- Containerized Docker deployment with OpenRouter cloud LLMs.
- Scalable Concurrency:** Handles multi-user customer showroom sessions simultaneously.
- Live Client Sharing:** Automatic temporary public URL generation via --share.

4. Innovation Pitch & Commercial Value Proposition

Transforming Luxury Sanitaryware Buying from Static Brochures into Interactive Spatial Intelligence

1. Hallucination-Proof Spatial CAD Engine • Eliminates mis-ordered fixtures, customer returns, and installation rework by deterministically validating room dimensions, 30" entrance door swings, and 15" fixture centerlines before quoting. • Recommends wall-hung fixtures for compact powder rooms and focal freestanding soaking tubs for luxury master suites.	2. 0ms Zero-Token Instant Intent Routing • 80% of customer interactions (greetings, catalog exploration, single-spec lookups) bypass LLM inference entirely via regex and keyword matching. • Saves 100% of LLM compute costs on routine queries and delivers instantaneous sub-millisecond response times for a snappy user experience.
3. Interactive 3D Spatial Studio Visualizer • Empowers homeowners and architects to visually manipulate room dimensions, set budgets, select aesthetic themes (Zen, Minimalist, Luxury, High-Tech), and view live 3D top-down layout diagrams. • One-click "Transfer to Advisor" instantly converts visual layouts into itemized commercial sales proposals.	4. Complete INR Commercial & Logistics Engine • End-to-end luxury sales workflow: itemized MRP pricing in Indian Rupees, automated 15% suite package discounts, exact 18% GST tax computations, and PIN-code delivery transit estimates in one unified proposal. • Eliminates mental arithmetic errors and delivers ready-to-purchase quotes.
COMMERCIAL ROI & SHOWROOM CONVERSION • Accelerated Sales Cycle: Shrinks design consultation and quoting turnaround from days to under 30 seconds. • Higher Average Order Value (AOV): Intelligent bundling increases multi-item package adoption across faucets, toilets, and showers. • Zero Installation Rework: Verified clearances eliminate expensive fixture return logistics.	PRODUCTION READINESS & GO-TO-MARKET • Zero-Setup Cold Start: Self-bootstrapping vector store indexes catalog automatically on launch without manual DB provisioning. • Full Test Automation: Verified against 12 end-to-end scenarios covering spatial constraints, math accuracy, and boundary checks. • Turnkey Deployment: One-click launch via run_demo.bat or Docker Compose.